

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0311

Commonwealth Chesapeake 345 kV
April 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0311. The Interconnection Customer proposes a 169.5 MW natural gas combined cycle facility interconnecting at the Commonwealth Chesapeake 345 kV substation in the Virginia area. The request was received on February 20, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 144 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,487,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tomstone-Bixby Hill 500.0 kV	Tombstone-Sharyland 345.0 kV	108.9	2,706	5.6
Sharyland-Tombstone 345.0 kV	Tombstone-Bixby Hill 500.0 kV	102.5	1,905	6.2
Sharyland-Bixby Hill 345.0 kV line	Tombstone-Bixby Hill 500.0 kV	102.4	1,851	7.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tomstone-Bixby Hill 500.0 kV	Reconductor line	\$71,690,000	1% (\$849,000)
Sharyland-Tombstone 345.0 kV	Reconductor line	\$16,300,000	2% (\$274,000)
Sharyland-Bixby Hill 345.0 kV line	Reconductor line	\$19,485,000	2% (\$364,000)
	Total allocated to project		\$1,487,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.